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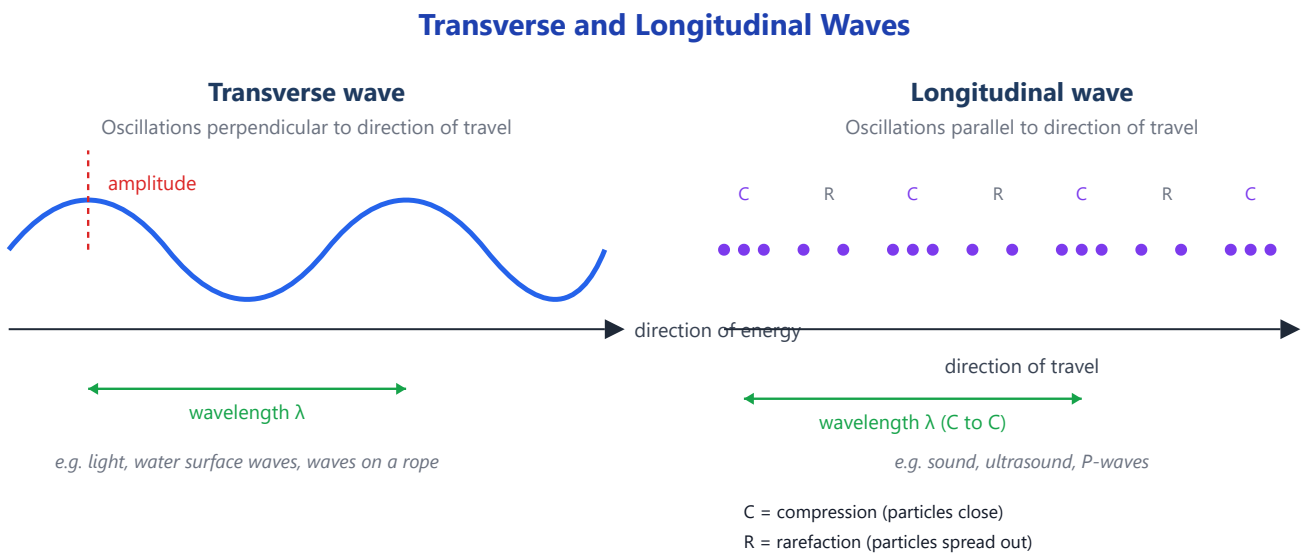
P1 — Radiation and Waves

OCR GCSE Physics B (J259) — 21st Century Science · Paper 1 (Breadth in Physics)

P1.1 What is a Wave?

> A **wave** is a disturbance that transfers **energy** (and information) from one place to another **without transferring matter**. The particles of the medium oscillate about their fixed positions; they do not travel with the wave.

Waves come in two main types based on the direction of the oscillation relative to the direction the wave is travelling.



Key Properties of a Wave

- **Amplitude (A)** — the maximum displacement of a particle from its rest position. Related to the *energy* the wave carries: bigger amplitude = more energy.

- **Wavelength (λ)** — the distance from one point on a wave to the same point on the next wave (e.g. peak to peak). Measured in metres.
- **Frequency (f)** — the number of complete waves passing a point per second. Measured in **hertz (Hz)**.
- **Period (T)** — the time for *one* complete wave cycle. `T = 1 / f`.
- **Wave speed (v)** — how fast the energy is transferred through the medium, in m/s.

$$\textbf{wave speed} = \textbf{frequency} \times \textbf{wavelength}$$

$$v = f \times \lambda \quad (\text{m/s}) = (\text{Hz}) \times (\text{m})$$

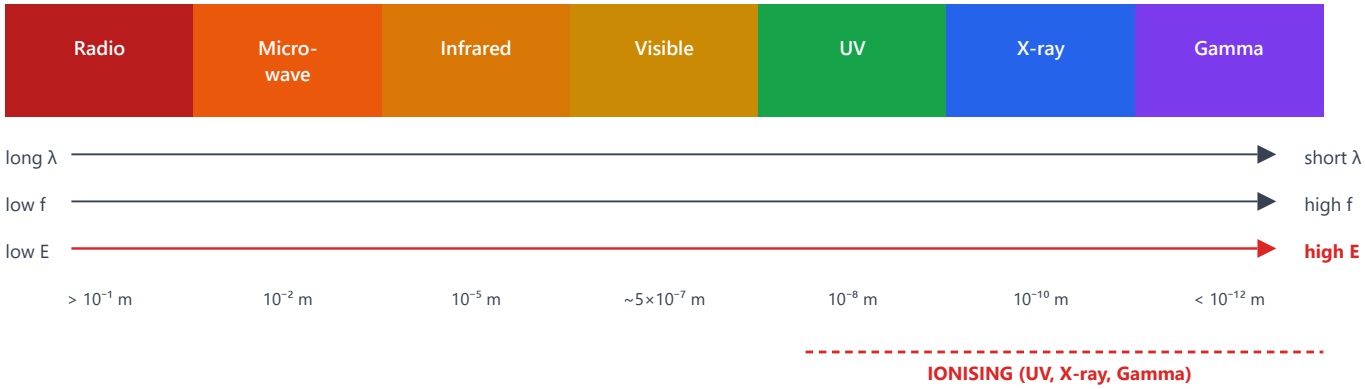
Worked Example: A sound wave has a frequency of 440 Hz and travels at 340 m/s in air. Find its wavelength.
λ = v / f = 340 / 440 = **0.77 m** (2 s.f.)

P1.2 The Electromagnetic Spectrum

All electromagnetic (EM) waves are **transverse**, travel at the same speed in a vacuum (**3×10^8 m/s**, the speed of light), and transfer energy by oscillating electric and magnetic fields — no medium is needed.

The **electromagnetic spectrum** is the full continuous range of EM waves, ordered from long wavelength / low frequency / low energy to short wavelength / high frequency / high energy.

The Electromagnetic Spectrum



Mnemonic: "Raging Martians Invaded Venus Using X-ray Guns"

Uses and Hazards of Each Part of the Spectrum

Type	Typical Use	Hazard
Radio	TV and radio broadcasting; long-distance communication; RFID tags	No significant hazard at low intensities
Microwave	Satellite communications (penetrate atmosphere), mobile phones, microwave ovens (absorbed by water)	Internal heating of body tissue at high intensity
Infrared (IR)	Thermal imaging, heaters, cooking, fibre-optic and short-range (TV remote) communication	Skin burns (absorbed as heat)
Visible light	Vision, photography, fibre optics, illumination	Extreme brightness can damage the retina (e.g. laser, Sun)
Ultraviolet (UV)	Fluorescent lamps, security marking, disinfecting water, sun-tanning	Skin cancer and premature ageing; eye damage
X-ray	Medical imaging of bones; airport security scanners; CT scans	Ionises cells → can cause cancer (dose must be minimised; radiographers shielded)

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Gamma	Sterilising surgical equipment and food; killing cancer cells (radiotherapy); medical tracers (PET, technetium)	Very penetrating; highly ionising; cell mutation and cancer risk

Exam Tip: A very common 6-mark question is to compare *two* parts of the spectrum — e.g. microwaves vs infrared for cooking, or X-ray vs gamma in medicine. Always link the **property** of the wave (wavelength, penetration, absorption) to **why** it suits the use.

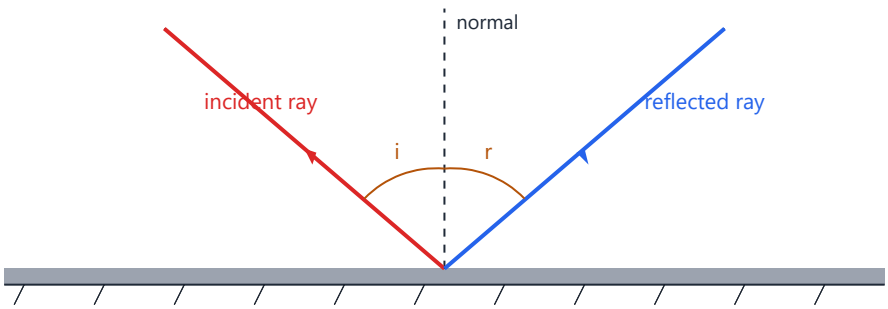
P1.3 Reflection, Refraction and Absorption

When a wave meets a boundary between two materials, one or more of the following can happen:

- **Reflection** — the wave bounces off the surface.
- **Refraction** — the wave changes direction as it crosses the boundary because its speed changes.
- **Absorption** — the wave's energy is transferred into the material (often as heat).
- **Transmission** — the wave passes through.

Reflection (Law of Reflection)

Law of Reflection



angle of incidence (i) = angle of reflection (r), both measured from the normal

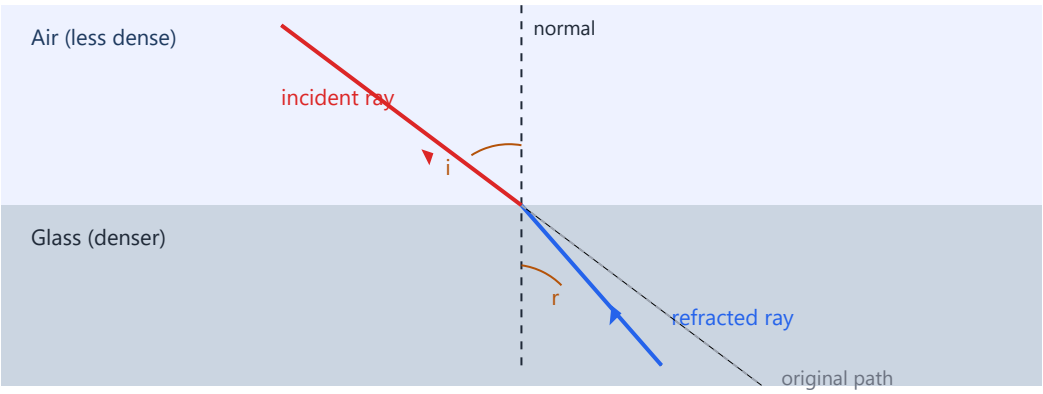
Refraction

When a wave passes from one medium into another its **speed changes**. Because the part of the wavefront in the new medium moves at a different speed to the part still in the old medium, the

wave **bends** (unless it hits the boundary head-on along the normal).

- Going into a **denser** (optically slower) medium: wave bends **towards** the normal.
- Going into a **less dense** medium: wave bends **away** from the normal.
- The **frequency does not change** during refraction — wavelength changes with speed.

Refraction from Air into Glass



> **Exam Tip:** Think of a toy car rolling off a smooth floor onto carpet: the wheel that hits the carpet first slows down, so the car turns *towards* the slower side. Same idea with light entering glass.

P1.4 Ionising vs Non-Ionising Radiation

> **Ionising radiation** has enough energy to knock electrons off atoms, creating charged ions. In living tissue, ionisation can damage DNA and cause cancer.

In the EM spectrum, the **ionising** waves are **ultraviolet, X-rays and gamma rays**. The lower-energy waves (radio, microwave, IR, visible) are **non-ionising**.

Radiation Dose (sievert, Sv)

The **equivalent dose** measures the risk of harm from ionising radiation, in **sieverts (Sv)**. 1 Sv is a very large dose; typical backgrounds are measured in millisieverts (mSv) or microsieverts (μSv).

Exposure	Typical dose
Dental X-ray	≈ 0.005 mSv
Chest X-ray	≈ 0.02 mSv
Transatlantic flight (cosmic rays)	≈ 0.08 mSv
UK annual background	≈ 2.7 mSv
CT scan (whole body)	≈ 10 mSv

Reducing Risk

- **Increase distance** from the source — intensity falls with distance squared.
- **Reduce exposure time**.
- **Use shielding** — lead aprons for X-rays; thick concrete walls around radiotherapy rooms; radiographers leave the room.
- **Use the lowest dose** that still gives a clear image / effective treatment.

P1.5 Waves in Different Media

The speed of a wave depends on the **medium**, not on its frequency.

Wave	Typical speed
Sound in air (20 °C)	≈ 340 m/s
Sound in water	≈ 1500 m/s
Sound in steel	≈ 5000 m/s
Light/EM in vacuum	3.00×10^8 m/s
Light in water	≈ 2.25×10^8 m/s

Light in glass	≈ 2.00×10^8 m/s

Sound travels faster through denser media because the particles are closer together and vibrations are passed on more quickly. Light, on the other hand, slows down in denser transparent materials.

P1.6 Sound and Ultrasound (overview)

Sound is a **longitudinal wave**. It needs a medium — it cannot travel through a vacuum. Humans typically hear 20 Hz to 20 000 Hz.

> **Ultrasound** is sound with a frequency above the human hearing range (> 20 kHz). It is partially reflected at boundaries between different media, which allows it to be used for imaging.

Uses of ultrasound:

- **Medical imaging** (pre-natal scans) — non-ionising, so safe for a foetus.
- **Industrial cleaning** and detection of flaws in metal castings.
- **Sonar** — measuring water depth; detecting shoals of fish or submarines.

> **Worked Example:** A sonar pulse returns 0.4 s after leaving a ship. Speed of sound in seawater is 1500 m/s. How deep is the seabed?
> Total distance = v × t = 1500 × 0.4 = 600 m. Depth = 600 / 2 = **300 m** (pulse travels there *and* back).

P1.7 Summary & Key Equations

- $v = f\lambda$
- $T = 1 / f$
- All EM waves travel at 3×10^8 m/s in a vacuum
- Energy increases with frequency: radio → gamma
- UV, X-ray, gamma are ionising

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Exam Tip: Never write "waves transfer matter". They transfer *energy*. This single-mark mark scheme point catches students out every year.